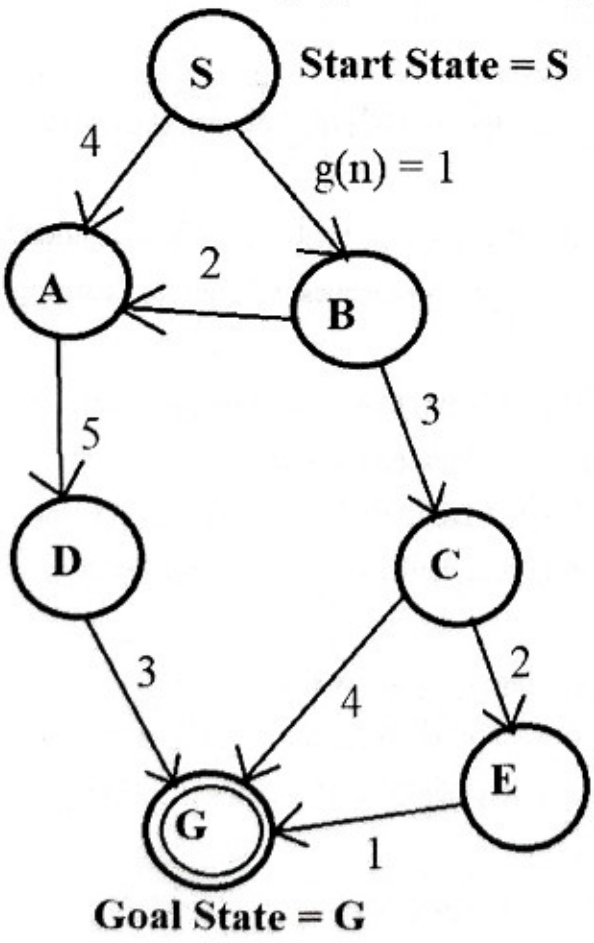
University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-IV, Odd Semester, Examination – 2024
Course: CSE-4131 (Artificial Intelligence)
(Session: 2020-2021 & 2019-2020)

Time: 3 Hours **Full Marks: 52.5**

[N. B. Answer any Six questions taking at least three questions from each section.]

SECTION-A

1. (a) Explain strong AI and weak AI with examples. 3.00
   (b) Define the following Terms: (i). Heuristic Knowledge, (ii) Structural Knowledge (iii) Meta Knowledge. 2.00
   (c) What are the main components of an AI system? Explain each. 3.75
2. (a) Explain the PEAS (Performance measure, Environment, Actuators, Sensors) description for AI agent. 3.00
   (b) Describe Model Based Reflex Agents with the basic diagram. Mention its applications. 3.75
   (c) What are the challenges arising to design agents for multi-agent environments? 2.00
3. (a) Explain the properties of the heuristic function. 2.00
   (b) Describe the step-by-step working principle of Best-First Search Algorithm. Write its disadvantages. 3.00
   (c) Consider the following Figure. Find out the goal path using A* search algorithm. 3.75



f(n) = g(n) + h(n)

State	h(n) Value
S	5
A	4
B	6
C	3
D	3
E	1
G	0

4. (a) Discuss the basic steps of resolution in AI. 3.00

(b) Consider the following facts: 3.00

(i) Bob likes all kinds of food.

(ii) Rice and Chapati are food.

(iii) Bob eats almond and is still alive.

(iv) Anything is eaten by anyone and still alive due to food.

Goal: Bob likes almond.

Prove it by resolution.

(c) Write the basic steps for conversion to the clausal form. 2.75

SECTION-B

5. (a) Draw a basic artificial neuron and explain its different parameters. ~~2.75~~ 3.25

(b) Explain the difference between shallow and deep neural networks. What is the role of an activation function in a neural network? ~~3.00~~ 3.50

(c) What is a loss function? Why is it necessary? 2.00

6. (a) Discuss the main components of a Convolutional Neural Network (CNN) and write its applications. 3.00

(b) What is the role of padding? Why do CNNs use the ReLU activation function? 2.00

(c) Define Transfer Learning. Describe common approaches of Transfer Learning with examples. 3.75

7. (a) Explain the structure of a DNN, including input, hidden, and output layers. What is the vanishing gradient problem in deep neural networks? 3.00

(b) Describe the basic steps to classify MNIST Digit classification using Convolution Neural Network (CNN). Show the graphical view of CNN steps for handwritten digit classification. 3.00

(c) Explain Max pooling and Average pooling for CNN with example. 2.75

8. (a) What is truth maintenance system? Give a proper example of the system. 1.75

(b) Define Inference Engine. Explain forward chaining and backward chaining with examples. 4.00

(c) Explain knowledge acquisition process in expert system with example. 3.00

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-4 Odd Semester Examination-2023
Course: CSE4131 (Artificial Intelligence)

Full Marks: 52.5 Duration: 3 Hours

[Answer any SIX questions taking THREE from each section]

SECTION: A

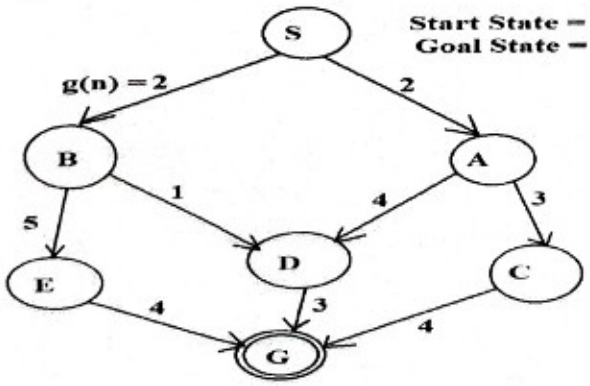
1. a) What is Intelligence? Explain different types of knowledge with example. 3
   b) Differentiate between declarative knowledge and procedural knowledge. 2
   c) Explain Knowledge Representation system in AI. Describe the basic components of knowledge Representation system with block diagram. 3.75

2. a) Explain basic types of agents in real world environment. What are the factors depending on Rationality of an agent? 3
   b) What are the basic structure of intelligent agents? Describe the structure of Learning based agents. 3.75
   c) Explain Turing test environment for artificial agents. 2

3. a) Define inference rules with example. Why is resolution required in AI? 2.75
   b) Describe the basic steps in Resolution. 2
   c) Consider the following knowledge based: 4
      i. Marcus was a man.
      ii. Marcus was a Pompeian.
      iii. All Pompeian's were Romans.
      iv. Caesar was a ruler.
      v. All Romans were either loyal to Caesar or hated him.
      vi. Everyone is loyal to someone.
      vii. People only try to assassinate rulers they are not loyal to.
      viii. Marcus tried to assassinate Caesar.
      GOAL: Does Marcus hate Caesar? Using resolution prove it.

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4. a) What is the blind search algorithm? Mention the applications of blind search algorithm. 2
   b) Discuss Depth-first search (DFS) algorithm with example. 3
   c) Consider the following Figure. Find out the goal path using A* search algorithm. 3.75



State	h(n) Value
S	7
A	5
B	4
C	6
D	8
E	6
G	0

SECTION: B

5. a) Why biasing is in artificial neural network. Explain excitatory and inhibitory weights in neural network. 2.5
   b) Consider a fully connected neural network 32-16-8-3 (Input layer-hidden layer-1 – hidden layer-2 – Output Layer). Draw the above neural network. Describe how many objects can be classified with this neural network. 3.25
   c) What is overfitting and underfitting in neural networks? Mention their drawbacks. 3

6. a) Discuss the application of fine tuning in Convolution Neural Network CNN. 2.5
   b) Explain the basic components of CNN. 2.5
   c) Define Transfer Learning. Describe common steps for Transfer Learning with examples. 3.75

7. a) State and explain Bayesian theorem for likelihood ratio. 4
   b) Describe three layer feed-forward Neural Network algorithm. 3
   c) Why hidden layers are used in multilayer Artificial Neural Network (ANN)? 1.75

8. a) Define and explain monotonic system and non-monotonic system with examples. 2.75
   b) Describe architecture of expert system. Write the functions of Inference Engine in expert system. 4
   c) Explain rule based expert system with example. 2

University of Rajshahi
Department of Computer Science and Engineering
B. Sc. Engg. Part-4 Even Semester, Examination-2022
CSE-4211 (Artificial Intelligence)

Time: 3 Hours **Full Marks: 52.5**

[Answer any **six** questions taking at least **three** from each Section]

Section-A

1. **(a)** Define artificial intelligence and describe its main goals. 3
(b) Explain the difference between strong AI and weak AI. Provide an example of each. 2.75
(c) Define a knowledge-based system and describe its main components. 3

2. **(a)** Define predicate logic. What are the logical notions of predicate logic? 3.5
(b) Explain first order predicate logic (FOPL). Describe the semantics of FOPL. 3.25
(c) Convert the following English sentences in FOPL: 2
- **i.** Some student in this class has visited Saint Martin's Island.
- **ii.** Every student likes gossiping.
- **iii.** None of friends are perfect.
- **iv.** All Romans were either loyal to Caesar or hated him.

3. **a)** Discuss the basic steps of resolution in AI. 3
b) Consider the following facts: 3.75
- **i.** Bread is a food.
- **ii.** Mango is a food.
- **iii.** Salma eats Burger.
- **iv.** Anything anyone eats is called food.
- **v.** Mili likes all kind of food.
- **vi.** Salma eats everything that Mili eats.
- **vii.** Does Mili likes Burger? Prove it by resolution.

c) Write the basic steps for conversion to clausal form. 2

4. **a)** Distinguish between blind search and directed search techniques. 2
b) Consider the 8-puzzle Problem. 2.75

Start State

2	8	3
1		4
7	6	5

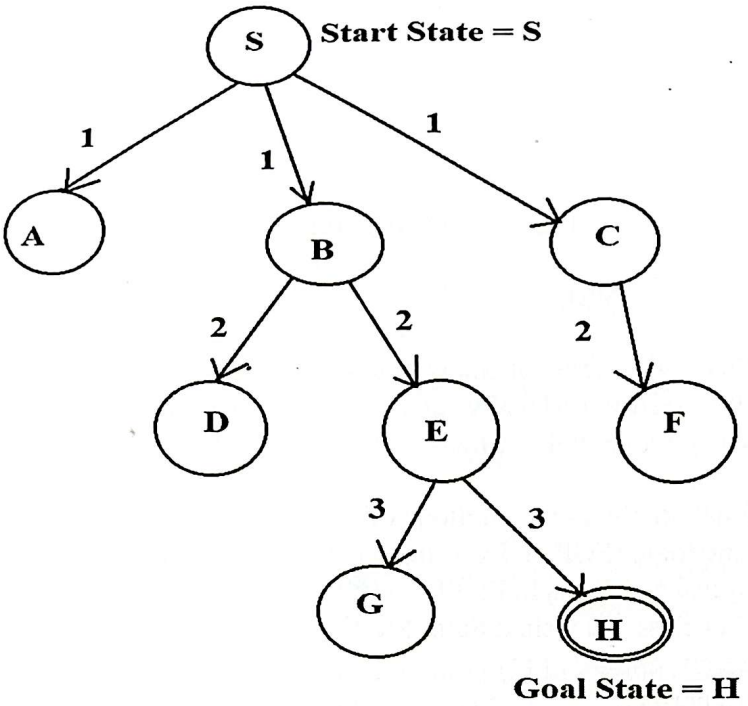
Goal State

1	2	3
7	8	4
	6	5

Find out the goal of above example using Heuristic search technique.

c) Consider the following figure. Find out the goal path using Best First Search algorithm with necessary explanation [h(n) = the heuristic value, g(n) = cost of reaching state n from the start state]. 4

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State	h(n)
S	12
A	14
B	11
C	10
D	8
E	6
F	7
G	4
H	0

Section-B

5. a) Describe monotonic system and non-monotonic system with examples. 2.5
   b) Define inference Engine. Explain forward chaining and backward chaining with examples. 3.75
   c) Explain Knowledge acquisition process in expert system with example. 2.5

6. a) Why activation function is used in neural network? Explain the following activation functions (i). ReLU activation function, (ii). Softmax activation function and (iii). Sigmoid activation function. 3
   b) Draw and explain fully connected neural network 8-16-8-4 (Input layer – Hidden layer-1- Hidden layer-2 – Output layer). Describe how many objects can be classified with this neural network. 3
   c) Differentiate between overfitting and underfitting in deep neural network (DNN). 2.75

7. a) Describe the importance of data augmentation and regularization techniques in training CNNs to improve generalization performance. 2.75
   b) Numerical Problem: Consider using a pre-trained VGG-16 model for a classification task with 10 classes. Assuming the last fully connected layer of the VGG-16 model has been replaced with a new fully connected layer for the task, compute the number of parameters in the new fully connected layer. 3
   c) Consider training a CNN for image classification using data augmentation with rotation, flipping, and scaling. If the original training dataset contains 1000 images and each image is augmented to create 5 additional variations, calculate the total number of augmented images in the dataset. 3

8. a) Explain crisp and non-crisp fuzzy logic. Write the applications of fuzzy logic. 3
   b) Discuss different types of Loss Functions for Artificial neural networks. 3.75
   c) Explain Reinforcement learning algorithm with diagram. 2

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University of Rajshahi

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.Sc. Engineering Part IV Even Semester Exam-2021

Course Code: CSE4211 **Course Title:** Artificial Intelligence

Time: 3 Hours Full Marks: 52.50

SECTION A

ANSWER ANY THREE OF THE FOLLOWINGS

1.(a) What is Artificial Intelligence (AI)? Describe the basic applications of AI. [2.00]

(b) Clarify knowledge and meta-knowledge with example. Differentiate between declarative and procedural knowledge. [3.00]

(c) Explain Knowledge Based System (KBS). Describe the components of a KBS. [3.75]

2.(a) Define artificial Agent. What is ideal rational Agent? [2.00]

(b) Describe the basic structure of goal based Agents. [3.75]

(c) Explain Turing test environment for artificial agents. [3.00]

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3.(a) Explain the basis of resolution. [2.00]

(b) Describe the resolution in predicate logic with example. [2.75]

(c) Assume the following facts: [4.00]

(i) Mr. X only likes easy courses.

(ii) Science courses are hard.

(iii) All the courses in the Finance department are easy.

(iv) F101 is a Finance course.

Use resolution to answer the question, "What course would Mr. X like?"

4.(a) Explain monotonic system with examples. [2.00]

(b) Describe architecture of expert system.Write the functions of Inference Engine in expert system. [3.75]

(c) Explain rule based expert system with example. [3.00]

SECTION B

ANSWER ANY THREE OF THE FOLLOWINGS

5.(a) What is Convolutional Neural Network (CNN)? [2.00]

(b) Explain the basic components of CNN. [2.25]

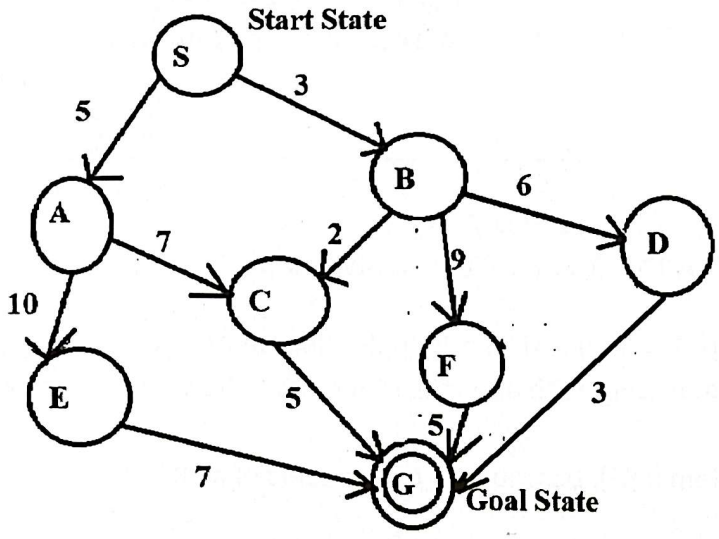
(c) Define Transfer Learning. Describe common approaches of Transfer Learning with examples. [4.50]

6.(a) Explain **Heuristic function**. Distinguish between Best-First search and A* search algorithms. [2.50]

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(b) Describe A* search Algorithm. [2.5

(c) Consider the following Figure. Find out the goal path using A* search algorithm. [3.75]



State No.	h(n) Value
S	5
A	10
B	4
C	5
D	7
E	8
F	4
G	0

7.(a) Describe the model of an artificial neuron. [2.00]

(b) Explain excitatory, inhibitory, and bias weights of neuron. [3.00]

(c) Consider the following three-layer backpropagation neural network with corresponding values inputs (x), weighted values (w), bias values (θ) = 0.5, the learning rate (η) =1 and Actual output = 1. Calculate the output for two times forward pass and one-time backward pass propagations. [3.75]

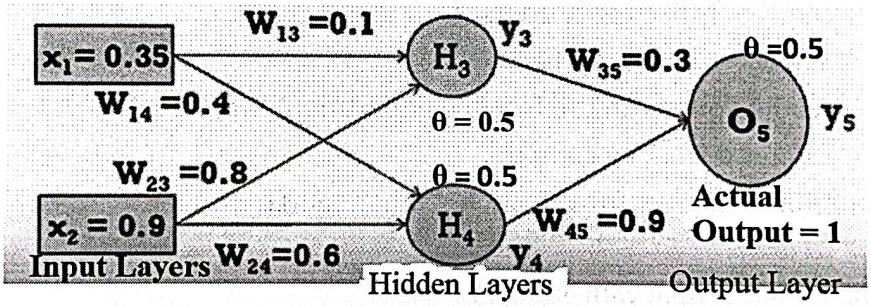


Figure: Three-Layer Backpropagation Neural Network

8.(a) Describe supervised, unsupervised and semi-supervised learning algorithms with example. [3.00]

(b) Why data augmentation is needed? Discuss data augmentation methods with example. [3.00]

(c) Discuss applications of TensorFlow and Keras in deep learning. [2.75]

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering Part IV Even Semester Examination 2020
Course Code: CSE 4211
Course Title: Artificial Intelligence

Full Marks: 52.5 **Time: 3 Hours**

[Answer any **SIX (06)** questions taking **THREE (03)** from each section]

Section A

1. a) Define Artificial Intelligence (AI). Explain the latest perception on AI. 2.75
   b) Distinguish between Human and Machine Intelligence. 3.00
   c) Explain Knowledge Base System (KBS). Describe the components of a KBS. 3.00

2. a) Define predicate logic. What are the logical notions in the predicate logic? 3.50
   b) Explain first order predicate logic (FOPL). Describe the semantics of FOPL. 3.25
   c) Express the following English sentences in FOPL: 2.00
      (i) All students like cricket.
      (ii) Some students are happy.
      (iii) Every teacher is liked by some students.
      (iv) Some real numbers are rational numbers.

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3. a) Explain resolution. Discuss basic algorithm of resolution. 3.00
   b) Consider the followingfacts: 3.75
      (i) All students who has been graduating are happy.
      (ii) All happy students smile.
      (iii) Someone has been graduating.
      (iv) Someone is smiling.

      We need to prove if someone is smiling. Using resolution find out goal.
   c) Find the meaning of the Statement = $(\neg PVQ)\&R \rightarrow SV(\neg R\&Q)$ for each of the interpretations given below: 2.00
      (a) I_1: P is true, Q is true, R is false, S is true.
      (b) I_2: P is true, Q is false, R is true, S is true.

4. a) Describe monotonic system and non-monotonic system with examples. 3.00
   b) Define inference Engine. Explain forward chaining and backward chaining with examples. 3.75
   c) Explain Knowledge acquisition process for expert system with example. 2.00

Section B

5. a) Explain the backtracking principle used in Turbo Prolog. 3.00
   b) What are the variable rules for Turbo Prolog? Explain anonymous variable with example. 3.00
   c) Write the format of **writef** predicate and **readint** predicate with example. 2.75

6. a) Define Heuristic search technique. Explain Heuristic function. 2.75
   b) Consider the 8-puzzle Problem. 3.00

Start

2	8	3
1	6	4
7		5

Goal

1	2	3
7	8	4
	6	5

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Find out the goal of above example using Heuristic search technique.

c) Describe Best-First Search (BFS) algorithm with suitable example. 3.00

7. a) Distinguish between blind search and directed search techniques. 2.00
   b) Describe the model of artificial neuron. What is function of hidden layer? 3.75
   c) Explain crisp and non-crisp fuzzy logic. Write the applications of fuzzy logic. 3.00

8. a) Why activation function is used in neural network? Explain at least three types of activation function. 3.00
   b) Draw and explain multilayer 64-8-4-3 neural network. Describe how many objects can be classified with this neural network. 3.75
   c) What is overfitting in neural network? 2.00

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engg. Part-IV, Even Semester, Examination 2019
Course: CSE-4211 (Artificial Intelligence)

Time: 3 Hours | Full Marks: 52.5

[N. B. Answer any Six questions taking at least three questions from each section.]

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SECTION-A

1.	(a)	Define Artificial Intelligence (AI). Explain latest perception on AI.	3
	(b)	Explain knowledge and meta-knowledge with example.	2.25
	(c)	Express the notion on reasoning. Write the basic goals of AI.	3.50
2.	(a)	Write down the steps to convert a FOPL sentence into clausal form.	4
	(b)	Convert the following expression to clausal form $\exists x\ \forall y(\ \forall z\ P(f(x), y, z) \rightarrow (\exists u\ Q(x, u)\ \&\ \exists v\ R(y, v)))$	3
	(c)	Prove that, "The sentence s is a logical consequence of a set of statement $\{s_1, s_2, \ldots\ldots, s_n\}$ if and only if $\{s_1\ \&\ s_2\ \&\ \ldots\ldots\ \&\ s_n\ \&\ \tilde{}s\}$ is inconsistent."	1.75
3.	(a)	Explain the properties of well-formed formulas (WFFs). How can you show that SUCCEED(bill) is a logical consequence? Thus, assume that both CLEVER(bill) and $\forall x$ CLEVER(x) $\rightarrow$ SUCCEED(x) are true under an interpretation.	1.50 + 2.50
	(b)	Consider the following clauses : i. GRANDFATHER(x, y) V ~FATHER(x, z) V ~PARENT(z, y) ii. PARENT(x, y) V ~ FATHER (x, y) iii. PARENT(x, y) V ~ MOTHER (x, y), iv. FATHER (Patterson, Patrick) v. FATHER (Patrick, Mike) By using resolution answer the following question, "Is Patterson a grandfather of Mike ?"	2
	(c)	Write the basic steps for conversion to clausal form.	2.75
4.	(a)	Explain monotonic system with examples.	3
	(b)	Describe architecture of expert system . Write the functions of Inference Engine in expert system.	3.75
	(c)	Explain rule based expert system with example.	2

SECTION-B

5.	(a)	Explain How prolog is different from procedural languages.	3
	(b)	Describe Prolog execution rule.	3
	(c)	Why 'fail' predicate is used? Explain with an example.	2.75
6.	(a)	Describe the basic structure of a prolog program.	2.5
	(b)	Write the basic rules for recursion in Prolog.	2.5
	(c)	Why dynamic database is used in prolog?	3.75
7.	(a)	Why searching technique is needed in AI? Explain Informed search technique.	2
	(b)	Describe the Hill Climbing searching algorithm with example.	4.75
	(c)	Explain crisp and non-crisp fuzzy logic. Write the applications of fuzzy logic.	2
8.	(a)	Define artificial neural network (ANN). Write the applications of ANN.	2
	(b)	Explain excitory, inhibitory and bias weights of neuron.	2
	(c)	Describe three layer feed-forward Neural Network algorithm.	3
	(d)	Distinguish between supervised and unsupervised learning technique.	1.75

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